

Internship Final Progress Report

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I. INTRODUCTION

In cooperation with the IBS Data Science Group, the KAIST Data Science Lab conducts several studies involving vast amount of data and various machine learning techniques. I joined the lab as an undergraduate intern for 3 months, from June to August during the summer of 2024.

II. THE DELTA PROJECT

The Delta project is an extension study following the development of Disaster Assessment with Vision foundation model (DAVI); an unsupervised change detection model using multi-temporal high-resolution satellite images to assess structural damage from a large-scale natural disaster. DAVI can also be used to calculate the speed of rebuilding collapsed buildings after the strike by performing change detection on multiple post-disaster time periods and observing the change in the damaged buildings.

This project suggests a potential use case of DAVI by demonstrating how the building recovery rate can be correlated with different socio-economic indicators. With Eiffel Ahn as a leader, the Delta project aims to propose using building damage recovery rate as a new national economic indicator.

In developing countries, it is often challenging to fund and conduct nation-wide research. Consequently, current economic indicators such as average wage, unemployment rate, and poverty rate may not accurately represent the entire population. Moreover, these data may be outdated or biased towards specific groups. Therefore, using the building recovery rate as a new economic indicator, as suggested by this study, could be more practical and efficient, especially in low-income countries. The Delta project thus seeks to provide a better method for assessing economic development using DAVI and satellite imagery.

III. DATA COLLECTION

The first step to calculate the building recovery rate is to detect the damaged buildings from the disaster event. The Delta Project is using DAVI to produce a binary mask of damaged buildings. Training DAVI with appropriate data is essential for its further application in calculating the building recovery rate. The Delta project utilizes DAVI to classify damaged buildings from their surroundings using satellite image pairs of the same area before and after a large-scale natural disaster. Therefore, the pre-disaster and

post-disaster satellite imagery pairs are required.

Labels of the damaged buildings are also necessary for training DAVI. These labels should be in the form of geometries or binary masks applied to broken or collapsed buildings. The process of creating these labels is time-consuming and inefficient, yet difficult to automate. Therefore, we sought matching satellite image pairs based on the building labels provided by online sources, such as Copernicus.

We focused on finding satellite images of developing countries, as the primary goal of the Delta project is to use the building recovery rate as a new indicator to address the issues posed by other economic indicators. We specifically targeted images of more developed areas, such as cities and towns, where building damages are more likely to be visible.

After examining various post-event satellite images from earthquakes, floods, wildfires, hurricanes, volcanic eruptions, and tsunamis, we decided to use images from before and after earthquakes, as the building damages were most apparent in these cases. Additionally, numerous images and statistical data are available online about earthquakes from several global-scale seismic activity monitoring systems, such as MAXAR, Copernicus, and ReliefWeb. These systems are implemented to prevent earthquake damage, given their potential destructiveness. We chose to use images from the Open Data Program Event Archive provided by Maxar Technologies.

A. Maxar Open Data

The Open Geospatial Solutions (opengeos) organization hosts a collection of open-source geospatial software projects. The projects are developed by a community of geospatial software developers and researchers, and are open-source licensed under the MIT license.

Maxar Technologies, a space technology company specializing in Earth observation and satellite products, provides the Maxar Open Data Program through opengeos. This program offers open-source pre- and post-event high-resolution satellite imagery to support emergency planning, risk assessment, monitoring of staging areas, emergency response, damage assessment, and recovery. The images are taken by the WorldView-2 or GeoEye-1 satellites, with each dataset in the catalog specifying the disaster event and the

affected country.

1) 2023 Turkey Earthquake: On February 6, 2023, a magnitude 7.8 earthquake struck southern and central Turkey, followed by a magnitude 7.7 aftershock later that day, centered in the city of Kahramanmaraş. The epicenter was about 24 km south of Kahramanmaraş and a similar distance northwest of Gaziantep, in south-central Turkey. Kahramanmaraş, the 10th largest city in Turkey, has a population of around 570,000. The earthquake's shallow focus, occurring only 10 km below the surface, made its impact particularly devastating.

More than 150,000 people were killed or injured, and according to Turkey's Disaster and Emergency Management Presidency, 280,000 buildings were severely damaged or reduced to rubble, with an additional 710,000 structures heavily damaged.

2) 2023 Morocco Earthquake: On September 8, 2023, a magnitude 6.8 earthquake struck Morocco, with its epicenter in the High Atlas Mountains, 71 km southwest of Marrakesh. The earthquake's focus, occurring at a depth of only 18 km below the surface according to the United States Geological Survey, produced stronger shaking at the surface than it would have had it been deeper. Marrakesh, with a population of 840,000, was the most impacted city.

As many as 10,000 people were killed or injured during the earthquake, with 59,000 homes damaged and 19,000 homes completely destroyed.

B. Image Pre-processing

To use the satellite imagery provided by the Maxar Open Data Program effectively, it needs to be pre-processed and modified into a format suitable for training DAVI with optimal performance. The images covering the same area during different time periods, including both pre-disaster and post-disaster, were selected to create image pairs.

Each satellite image was taken at different times and at different angles relative to the Earth's surface, so we had to ensure their usability before downloading. The image selection process involved visually checking each image using an image visualization program on Jupyter Notebook provided by the Maxar Open Data Program. The main criteria for a good image were:

- high image resolution - Ground Sample Distance (GSD) < 0.50 m,
- minimal cloud coverage - area cloud coverage < 10%
- off nadir angle < 30°

For estimating cloud coverage, we only considered the areas where labels existed. As a result, a few selected images had some cloud coverage in unwanted regions, and these were manually removed.

After checking the images, the IDs of the pre- and post-event satellite imagery pairs were recorded. These recorded ID pairs will be used to align the cropped images in the future.

Due to the limited availability of pre- and post-event image pairs, we decided to include pairs of pre-disaster images taken at different times before the disaster. These image pairs were treated in the same way during the data pre-processing stage.

1) Downloading Satellite Images: For each disaster event, a set of GeoJSON files for multi-spectral satellite images before and after the disaster were provided. These files were used to download raw image footprints in TIFF format. The images were downloaded using these files through the maxar-open-data module of the leafmap Python package.

2) Tiling Images: While experimenting with DAVI, we found that it produced the best change detection results when using 256 x 256 pixel images at zoom level 18. Therefore, I used the generate_tiles method from the gdal2tiles Python library to alter the zoom level of the images and crop them to the desired size and zoom level. As this function uses the Open Source Geospatial (OSGeo) Tile Map Service Specification and EPSG:4326 projection when generating tiles, every satellite image was cropped using the same base tiles. Thus, the generated tiles from previously recorded image pairs and the building labels could be easily aligned. All cropped images were stored in a directory with the image ID of the original satellite imagery.

3) Producing Image Pairs: The longitude and latitude of the tiled image files were calculated from the metadata of each tile. These coordinates were then used to find matching pairs of pre-disaster and post-disaster images. By examining each cropped image in one of the image IDs from a pair, the pre-processing program searched for an image with the same longitude and latitude coordinates in the matching image ID directory. Once a pair of cropped images was found, they were copied and stored in a separate folder along with all other cropped image pairs.

4) Pruning Images: To ensure a good performance of the model, it was crucial to remove images that might cause difficulties while training DAVI. Images that might be misleading, do not clearly display buildings, or could negatively impact DAVI's performance were removed. Specifically, I removed images that:

- contain empty pixels - if the image is from the edge of the original satellite imagery and the base tile does not align perfectly,
- have high cloud coverage,
- only cover mountainous or plain land without any built structures, or
- are distorted, making the buildings unrecognizable.

Both images in a pair were deleted if either one was

removed after this image pruning process.

IV. BASELINE RESEARCH

To evaluate the performance of DAVI when detecting damaged buildings, we will be using two other unsupervised CD methods for comparison.

A. Deep Change Vector Analysis (DCVA)

DCVA [SBB19] is an unsupervised method designed for multiple CD in Very High-Resolution (VHR) satellite images. The model generates high-level feature representations of the images using pretrained CNNs. These features are then compared through Change Vector Analysis (CVA), which computes a change vector by calculating the difference between the feature vectors of corresponding pixels in two images. The resulting change vectors are analyzed to identify areas where significant changes have occurred.

DCVA focuses on capturing changes within images, making it effective for general change detection tasks. However, it is less adaptable and struggles to perform disaster-specific assessments across different types of disasters, when compared with DAVI.

B. KPCA Convolutional Mapping Network (KPCA-MNet)

KPCA-MNet [Wu+22] uses a deep siamese network consisting of weight-shared KPCA convolutional layers to extract feature maps from multitemporal VHR satellite images. Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms original data into a new feature space while preserving most of the original information. Kernel PCA (KPCA) is a nonlinear extension of PCA, mapping the original images into a new nonlinear feature space with higher dimensionality. KPCA can extract representative features similar to CNNs, but its learning process is unsupervised, hence it does not require annotated data. After feature extraction, a feature difference map is created through pixelwise subtraction. The difference features are then mapped into a 2D polar domain, followed by threshold segmentation to generate a binary change detection result.

The biggest difference between DAVI and KPCAMNet lies in the feature extraction process. DAVI utilizes pretrained CNN models, while KPCAMNet relies on the KPCA method to extract features. KPCAMNet's approach allows it to detect intricate patterns and improve the representativeness of features. However, DAVI excels in understanding image content more deeply and generalizing across diverse tasks due to its use of pretrained CNNs.

Thus, KPCAMNet focuses primarily on detecting changes based on pixel and feature-level differences without a deep understanding of the scene's context or semantics. In

contrast, DAVI, by leveraging pretrained CNN models, incorporates both contextual and semantic understanding, enabling it to better distinguish between normal variations in multitemporal images (e.g., seasonal changes or shadows) and disaster-induced changes. This ultimately enhances DAVI's accuracy in CD, particularly for disaster response applications.

However, while implementing the KPCAMNet model using the provided GitHub code, I encountered an issue where the model took an excessively long time to train. This is expected as the KPCA technique is known to be computationally intensive. Hence, I was unable to implement the model for comparison with DAVI for the study.

V. CONCLUSION

During my internship at the KAIST Data Science Lab, I had the opportunity to contribute to the Delta Project to explore the potential of DAVI for disaster recovery assessment. The use of the building recovery rate as a new socio-economic indicator could provide a more accurate and timely measure of economic resilience and recovery, especially in developing countries.

By focusing on satellite imagery from earthquake events in regions like Turkey and Morocco, the next step would be to apply DAVI to real-world scenarios, identifying areas of significant structural damage and tracking recovery over time. To train DAVI for tracking the building recovery rate, further efforts are needed to acquire building labels and masks of structural damage from disaster events.

Overall, this experience taught me how to pre-process data for training deep learning models and developed my ability to effectively read, understand, and evaluate the studies on research papers. It also helped deepen my understanding of data science applications in real-world contexts and highlighted the importance of using appropriate data to optimize models and derive valuable insights.

REFERENCES

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